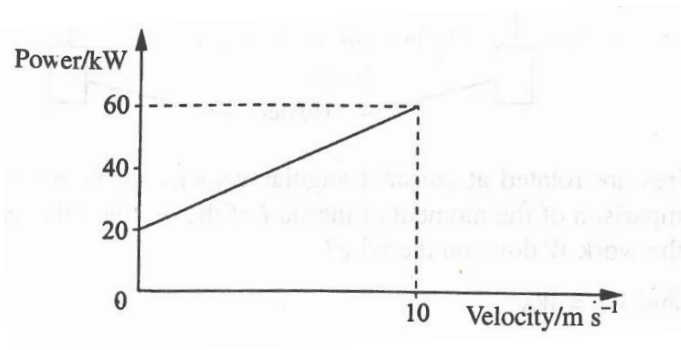


- 8 The power of a motorcycle varies with its velocity as shown in the figure below.



If the total mass of the motorcycle and its rider is 500 kg, what is the acceleration of the motorcycle?

- A** 1.25 m s⁻² **B** 4.00 m s⁻² **C** 8.00 m s⁻² **D** 12.5 m s⁻²

Ans: C

$$P = P_0 + Fv$$

7

$$F = (60-20) \times 10^3 / (10-0) = 4 \times 10^3 \text{ N}$$

$$a = \frac{F}{m} = \frac{4 \times 10^3}{500} = 8 \text{ ms}^{-1}$$